AI Interviewer Agent Using Python

IEEE Style Mini Project Report

Student: Purnima Alla
Roll No: 23U41A4403
Branch: CSDM
DADI Institute of Engineering

Abstract

This project presents an AI Interviewer Agent developed using Python. The system automates the initial interview process by asking predefined technical questions, accepting candidate responses, evaluating them through keyword matching, calculating scores, and generating feedback. The solution demonstrates the application of Artificial Intelligence concepts in recruitment automation while remaining simple enough for academic learning.

Keywords

Artificial Intelligence, Python, Interview Automation, Natural Language Processing, Recruitment.

1. Introduction

Recruitment is an essential process in every organization. Manual interviews require considerable time and effort. AI-powered interviewer agents help automate the initial screening process, ensuring consistency and reducing human workload. This mini project demonstrates how Python can be used to build a basic interview agent that evaluates answers and provides instant feedback.

2. Literature Survey

Several organizations use AI-based interview systems to perform preliminary candidate assessments. Modern platforms employ Natural Language Processing, Machine Learning, and Large Language Models to understand responses. This project uses a lightweight keyword-based evaluation approach suitable for educational purposes.

3. Objectives

• Automate interview questioning.
• Evaluate candidate responses.
• Calculate interview scores.
• Generate instant feedback.
• Demonstrate AI concepts using Python.

4. Proposed Methodology

The system contains a Question Bank, Answer Collection Module, Evaluation Engine, Score Calculator, and Feedback Generator. The candidate enters answers through the console. The program compares responses with predefined keywords and awards marks accordingly.

5. Algorithm

Step 1: Store interview questions.
Step 2: Accept candidate answers.
Step 3: Convert answers to lowercase.
Step 4: Compare with expected keywords.
Step 5: Update score.
Step 6: Display final result and feedback.

6. Python Code

```
questions={
 'What is Python?':['language','programming'],
 'What is AI?':['machine','intelligence'],
 'What is OOP?':['object','class']
}
score=0
for q,keys in questions.items():
 ans=input(q+' ').lower()
 if any(k in ans for k in keys):
  score+=1
print('Score:',score,'/',len(questions))
if score==3:
 print('Excellent')
elif score==2:
 print('Good')
else:
 print('Needs Improvement')
```

7. Sample Output

```
What is Python? Programming language
What is AI? Machine intelligence
What is OOP? Object-oriented programming using classes
Score: 3/3
Excellent
```

8. Advantages

The system reduces interviewer workload, provides consistent evaluation, gives instant feedback, is easy to implement, and can be extended with speech recognition and large language models.

9. Applications

Campus recruitment, HR screening, mock interviews, placement preparation, and online education.

10. Future Scope

Future improvements include voice-based interviews, resume analysis, emotion detection, cloud deployment, integration with LLMs, and multilingual support.

11. Conclusion

The AI Interviewer Agent demonstrates how Artificial Intelligence concepts can automate the interview process using Python. Although simple, the project provides a strong foundation for advanced AI recruitment systems.

References

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